

Financial Data Trust & Market Stability Intelligence Platform

Real-Time Cybersecurity Layer for Market Data
Integrity & Systemic Risk Detection

Presented by
Team Synergy

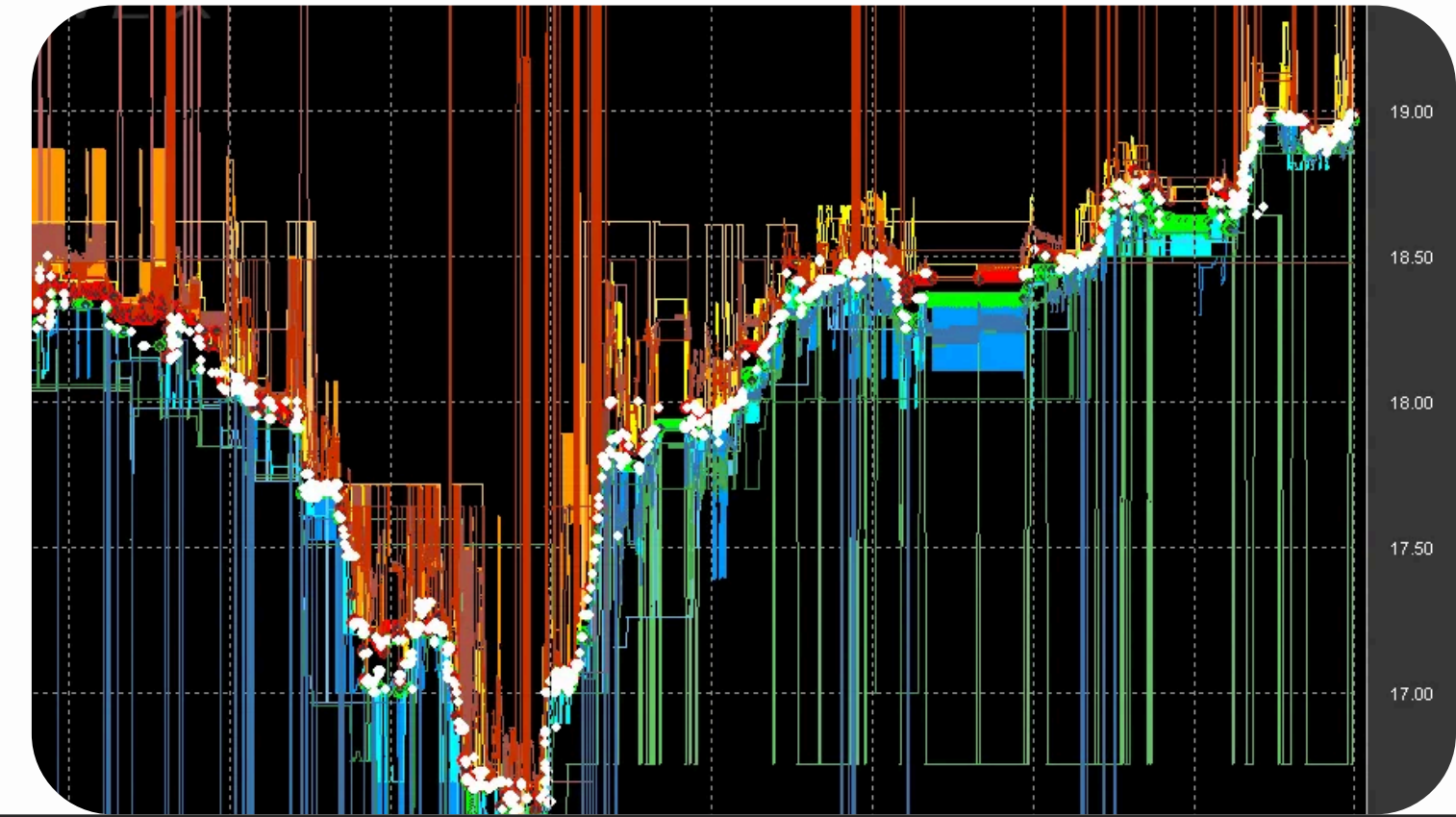
Problem Statement

Financial markets are fully automated.

Algorithms trust incoming market data blindly.
If data is corrupted or abnormal → automated losses.
If multiple assets destabilize → systemic risk.

There is no unified system combining:

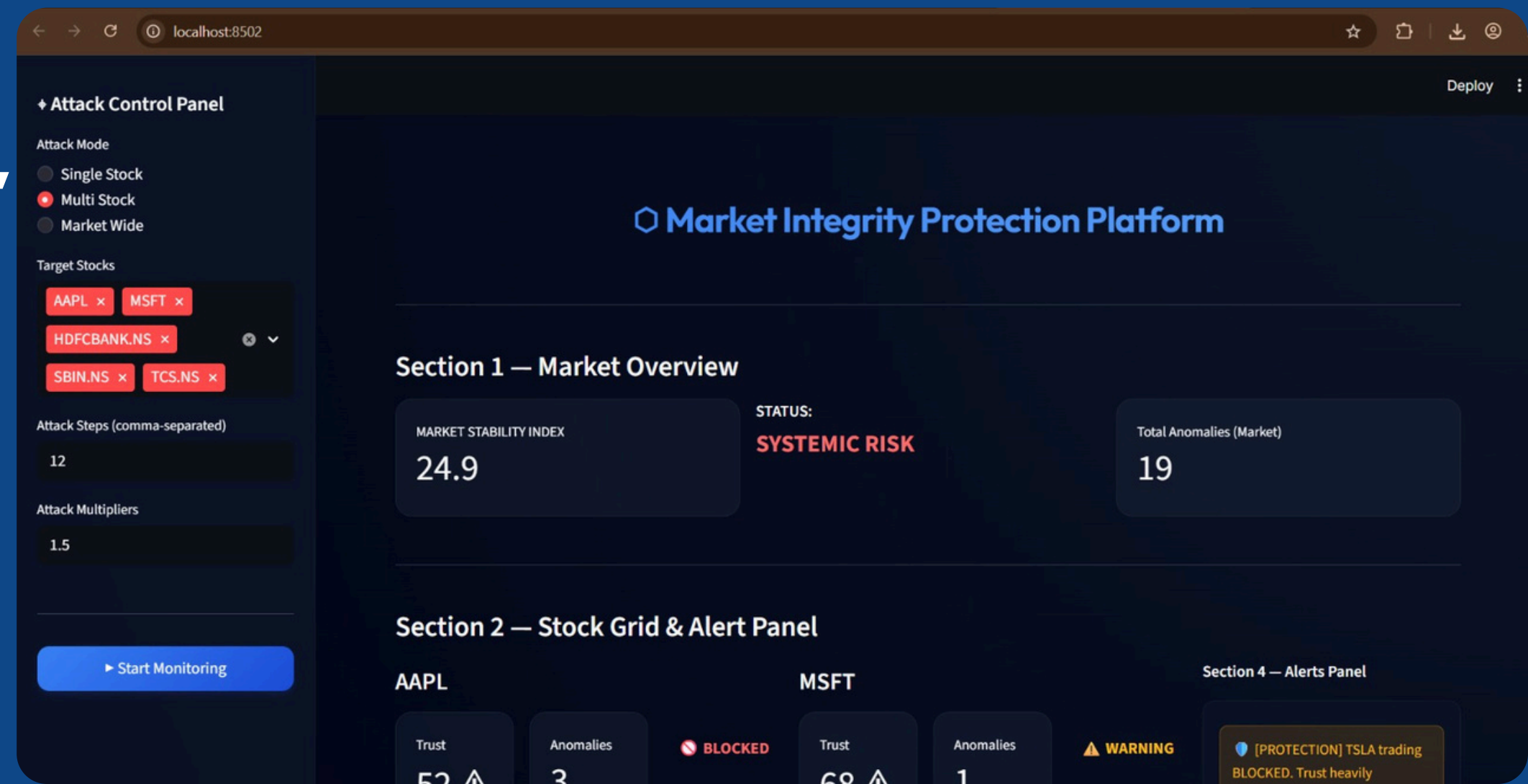
- Data validation
- Anomaly detection
- Contagion monitoring
- Resilience testing



Our Solution

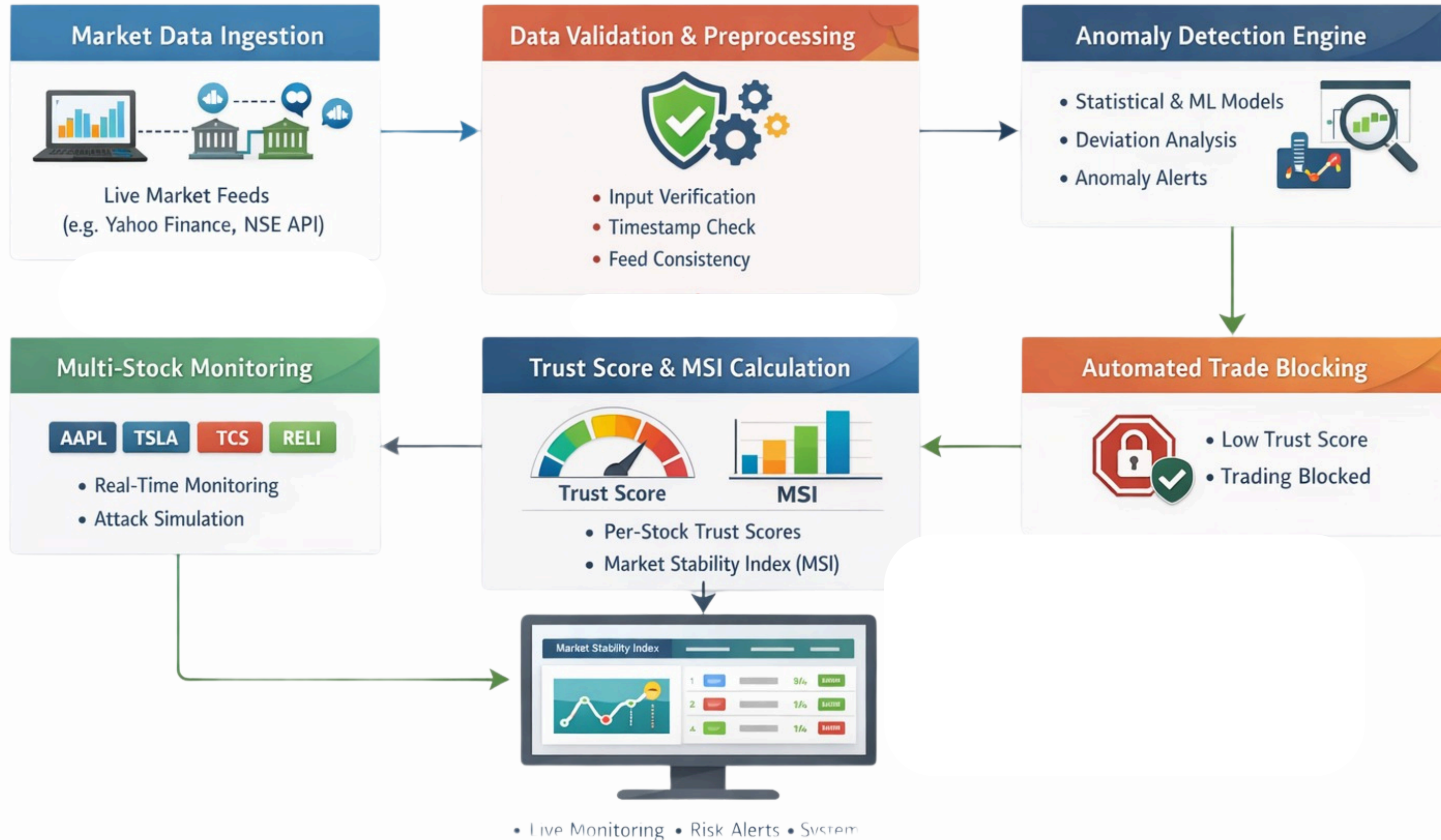
- We built a system that continuously monitors stock market data from two independent sources, detects abnormal behavior in real time, measures overall market stability, and automatically flags or blocks risky trading activity before it causes larger damage.

**Real-time market
Integrity +
Systemic Risk
Intelligence in one
unified cyber-
financial engine.**

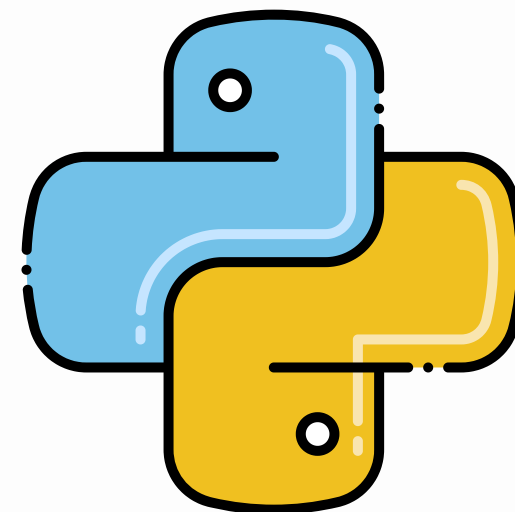


Workflow Overview

$Z = \text{current price} - \text{Mean}/\text{SD}$
 $\text{Severity} = \min(|Z|/\text{threshold}, 1.0)$
 $\text{risk memory} = \text{severity} + (1-z) \times \text{prev risk}$
 $\text{Trust score} = 100 \times (1 - \text{risk memory})$



Technology Stack



Target Users

01

High-Frequency Trading Firms

Protect automated trading strategies by ensuring secure, fast, and reliable execution systems that minimize risk and prevent market manipulation.

02

Stock Exchanges

Maintain internal feed integrity monitoring to ensure accurate price dissemination and prevent data tampering or disruptions.

03

Regulators

Enable early detection of systemic instability through continuous monitoring and risk assessment mechanisms.

Novelty :

First platform combining dual-feed data validation, attack simulation, and systemic market stability scoring in real time.

Business : \$60–75 B

TAM

\$8–12 B

SAM

\$50M–150M

SOM

Societal Relevance & Impact

- **Protecting Public Wealth**

Safeguards citizens' investments, savings, and retirement funds

- **Economic Stability**

Reduces market disruptions affecting national economy

- **Digital Financial Trust**

Enhances transparency and confidence in digital markets

- **Regulatory Support**

Enables proactive risk monitoring and governance

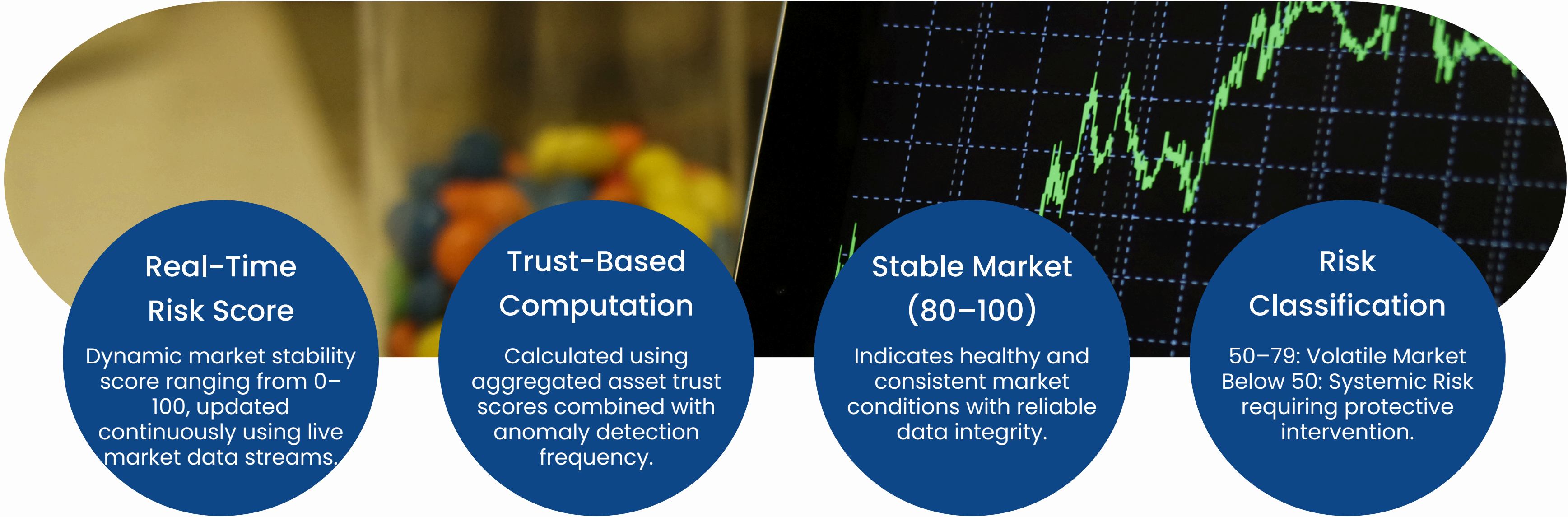
- **Resilient Financial Future**

Promotes stable, secure, and innovation-driven markets



Core Innovation: Market Stability Index (MSI)

Real-Time Market Risk Intelligence



Real-Time Risk Score

Dynamic market stability score ranging from 0–100, updated continuously using live market data streams.

Trust-Based Computation

Calculated using aggregated asset trust scores combined with anomaly detection frequency.

Stable Market (80–100)

Indicates healthy and consistent market conditions with reliable data integrity.

Risk Classification

50–79: Volatile Market
Below 50: Systemic Risk
requiring protective
intervention.

THANK YOU

ANY QUERIES?

HIGH FREQUENCY TRADING (HFT)



HEDGE FUNDS



Exchanges (Infrastructure-Level Targets)

1. NSE (National Stock Exchange – India)
2. BSE (Bombay Stock Exchange)

3. NYSE
4. NASDAQ